



# Joel Oskarsson

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ETH Zurich AI Center, Zurich, Switzerland

## Academic Career

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2025 – **Postdoctoral Fellow, ETH Zurich AI Center**, Zurich, Switzerland

- Awarded 2-year fellowship in highly competitive call with a **4% acceptance rate**.
- Mentored by Torsten Hoefler (Computer Science), Siddhartha Mishra (Math), and Benedikt Soja (Space Geodesy). Topic: *Modeling the Earth System in Latent Space*.
- Funded by the Swiss AI Initiative, part of effort to develop **AI foundation models**.

2020–2025 **PhD in Computer Science, Linköping University**, 240 ECTS

- Thesis: *Modeling Spatio-Temporal Systems with Graph-based Machine Learning*
- Supervised by Fredrik Lindsten, Per Sidén, Tomas Landelius.
- Affiliated with WASP graduate school.

2015–2020 **Master's in Computer Science and Engineering, Linköping University**, 300 ECTS

- **Exchange year at ETH Zurich**, Zurich, Switzerland (2018–2019).

## Research

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### *Summary of publications*

*See separate document for full list of publications.*

- 5 conference papers, 2 journal papers, 5 workshop papers, + multiple works under review.
- 285 total citations, h-index 9 (from Google Scholar).

### *Software Research Output*

Creator and maintainer of the *Neural-LAM* **open source library** for AI weather forecasting research.

- Translates research findings into direct community impact through open-source tooling.
- **Over 250 GitHub forks and stars**, reflecting widespread usage.
- Used by multiple **international research groups**, basis for many peer-reviewed publications.
- Organizational participant in Google Summer of Code 2026, personally acting as project mentor.

### *Invited Talks*

2025 **Google DeepMind**

*Probabilistic Weather Forecasting with Hierarchical Graph Neural Networks*

2024 **RISE Learning Machines Seminars**

*Frontiers in Machine Learning for Weather Forecasting*

**European Centre for Medium-Range Weather Forecasts (ECMWF)**

*Graph-based Machine Learning for Weather*

**University of New South Wales (UNSW)**

*Graph-based Machine Learning for Spatio-Temporal Data, with Application to Traffic and Weather Forecasting*

**Uppsala University**

*Graph-based Neural Weather Prediction for Limited Area Modeling*

Further invited talks: WeatherGenerator Dissemination Workshop (2026); MeteoSwiss (2024); ETH Zurich (2024); IEA Wind Task 51 Webinar: Forecasting for the Weather Driven Energy System (2024); Danish Meteorological Institute (2023).

## ***Presentations at Scientific Events***

- Presentations at top **machine learning conferences**, and accompanying workshops: NeurIPS (2024: poster; 2023: workshop poster), ICML (2024: workshop poster), AISTATS (2023: poster).
- Selected for **oral presentation** at the Machine Learning for Earth System Modelling Workshop, Bonn, Germany, in **three consecutive years** (2023, 2024, 2025).
- Oral presentation at the KVA Symposium on AI for Science, Stockholm (2025)
- Oral presentation, poster and panel discussion at the ESA-ECMWF Workshop on Machine Learning for Earth Observation, Frascati, Italy (2024)

Selected further presentations: Climate Informatics, Lausanne, Switzerland (2026, oral); Learning on Graphs and Geometry Meetup, Uppsala (2025, oral); NORDSTAT, Gothenburg (2023, oral); Swedish Symposium on Image Analysis and Deep Learning, Norrköping (2023, poster);

## ***Research Visits***

2023 Five-week research visit to **University College London (UCL)**, London, U.K.

Hosted by Prof. Marc Deisenroth, Sustainability and Machine Learning Group.

Shorter visits: Norwegian Meteorological Institute, Oslo (2026); ECMWF, Bonn (2024); ETH Zurich + MeteoSwiss, Zurich (2024); Finnish Center for AI + Aalto University, Helsinki (2022, WASP International Study Visit).

## ***Career and Compute Grants***

- Research Visit Scholarship ( $\approx 60,000$  SEK), WASP Program.
- Co-author of applications for compute resources from NSC, Sweden (awarded) and CSCS, Switzerland (under review).

## **Teaching, Supervision and Mentoring**

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- Supervisor of **8 master's thesis projects** (LiU, 2021–2024) and **5 ongoing projects** (ETHZ 2026).
- During my PhD and as postdoctoral researcher, I have mentored junior PhD students through collaborative research, co-authorship, and project guidance.
- Co- and guest-lecturer for courses *AI Center Projects in Machine Learning Research* (ETHZ) and *Time Series and Sequence Learning* (LiU).
- **Co-course-developer** of the *Foundations of Machine Learning* online course (LiU, 2021–2024).
- Pedagogical training: *Becoming a Teacher in Higher Education*, 6 ECTS (LiU, 2021).

Other teaching assistant work (2016–2024): Advanced Machine Learning (5 $\times$ ), Computational Statistics, Foundations of ML, Building AI, Introduction to Python, etc. All at LiU.

## **Media Coverage and Scientific Outreach**

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- *De vill göra väderprognoser mer pålitliga – med hjälp av AI*, SVT (2024): **Television news feature**.
- *Förbättrade väderprognoser med AI*, Linköping University (2023): News article on University website.
- *Forskningsframsteg inom Maskininlärning för Väderprognoser* (2025): **Popular science article** in Polarfront, journal of the Swedish meteorological society.
- *AI för prognosmeteorologer* (2025): Lecture for forecast meteorologists about AI weather forecasting.

## **Service and Skills**

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### ***Peer Reviewing***

NeurIPS (2024); ICLR (2025, 2026); AISTATS (2023, 2024, 2026); Climate Informatics (2026); Geophysical Research Letters (2025); SOLA (2025); Geomatics, Natural Hazards and Risk (2025, 2026); CCAI workshop (2024, 2025).

### ***Other Service***

2025–2026 Reviewer for applications to ETH Zurich AI Center Fellowship 2026.

2023–2025 Co-organizer of monthly online meetups for ML limited area weather forecasting researchers.

2022–2025 Organizer of the *STIMA ML Reading Group*, Linköping University.

### ***Skills***

- Accustomed to high-performance computing systems and linux environments.
- Languages: Swedish (native), English (fluent), German (basic).